

OMOP Database for Toxicology Terminology

- **Concept Table:**

- Toxicology substances
- Categories
- Subcategories

concept id	concept name	concept code
2,000,110,223	6-ACETYLCODEINE	substancelid

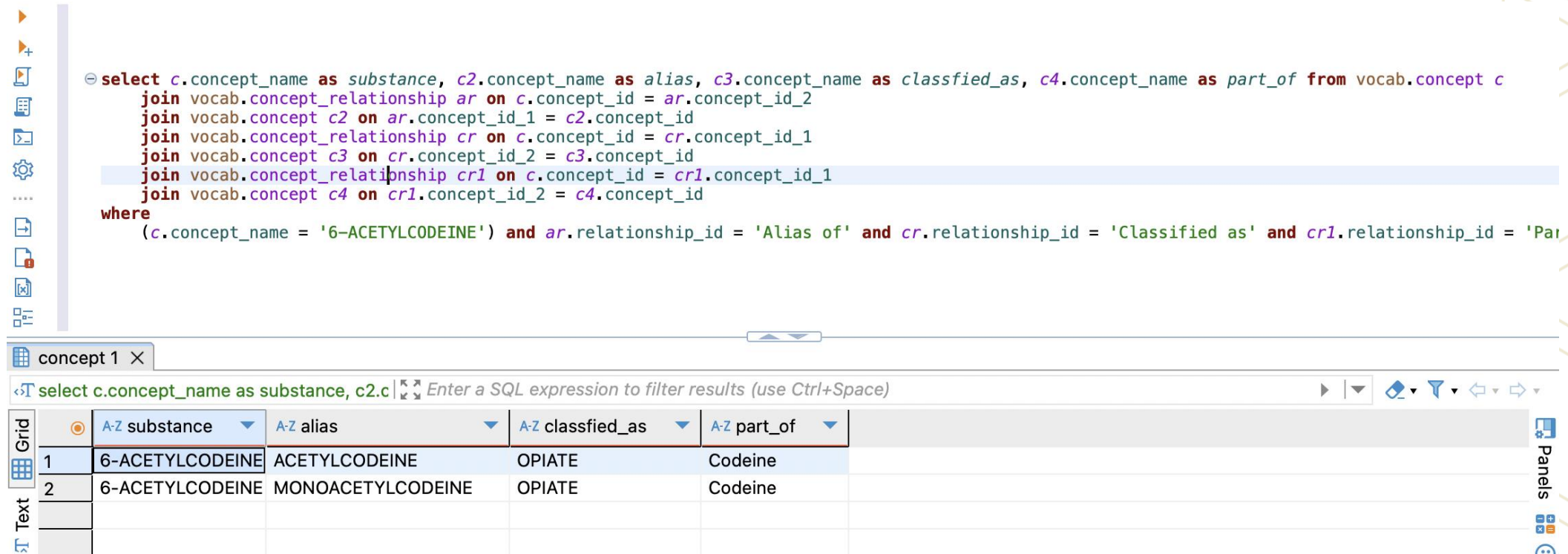
- **Relationship Table:**

- “Alias of” for synonyms
- “Classified as” for categories
- “Part of” for subcategories
- More relationships can be added

- **Concept Relationship Table:**

- Substances with relationship(s)
- Currently substances loaded with Alias, Categories, and Subcategories

Substance Example in OMOP DB



The screenshot shows a web-based interface for querying an OMOP database. The top section displays a SQL query in a text editor. The query selects concept names and relationships for a specific substance (6-ACETYLCODEINE). The bottom section shows the results in a table grid.

```
select c.concept_name as substance, c2.concept_name as alias, c3.concept_name as classified_as, c4.concept_name as part_of from vocab.concept c
join vocab.concept_relationship ar on c.concept_id = ar.concept_id_2
join vocab.concept c2 on ar.concept_id_1 = c2.concept_id
join vocab.concept_relationship cr on c.concept_id = cr.concept_id_1
join vocab.concept c3 on cr.concept_id_2 = c3.concept_id
join vocab.concept_relationship cr1 on c.concept_id = cr1.concept_id_1
join vocab.concept c4 on cr1.concept_id_2 = c4.concept_id
where
(c.concept_name = '6-ACETYLCODEINE') and ar.relationship_id = 'Alias of' and cr.relationship_id = 'Classified as' and cr1.relationship_id = 'Part of'
```

concept 1 X

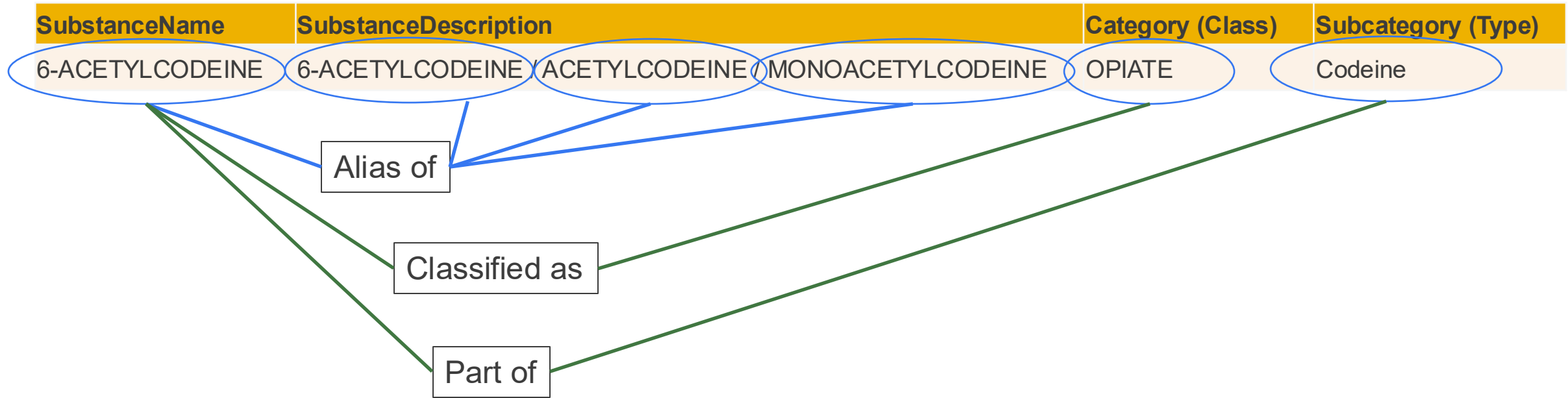
select c.concept_name as substance, c2.c Enter a SQL expression to filter results (use Ctrl+Space)

	A-Z substance	A-Z alias	A-Z classified_as	A-Z part_of
1	6-ACETYLCODEINE	ACETYLCODEINE	OPIATE	Codeine
2	6-ACETYLCODEINE	MONOACETYLCODEINE	OPIATE	Codeine

UI based tool is available from Vocab Mapping project (requires some modifications).

- Users won't need to do SQL

Substance Example in OMOP



This proof of concept was developed based on the active substances list available in the National Violent Death Reporting System (NVDRS)/State Unintentional Drug Overdose Reporting System (SUDORS)

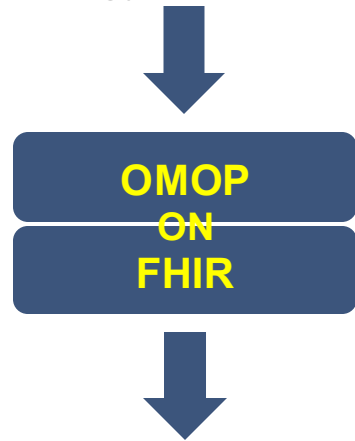
- ToxSubstance
- ToxCategory
- ToxSubcategory

Preferred Substance Name Search using FHIR API

- API URL: [http://<host>/ConceptMap/\\$translate](http://<host>/ConceptMap/$translate) with the following input parameters

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "coding",
      "valueCoding": {
        "system": "ToxSubstance ",
        "display": "ACETYLCODEINE"
      }
    }
  ]
}
```

Toxicology Lookup Table



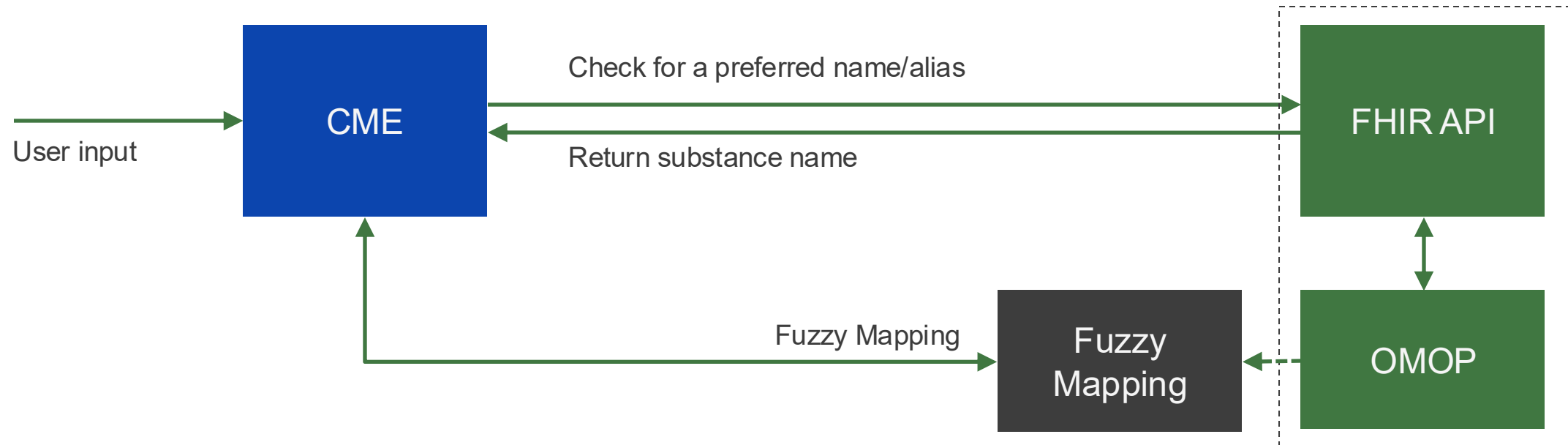
FHIR ConceptMap API

- Response:

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueBoolean": true
    },
    {
      "name": "match",
      "part": [
        {
          "name": "equivalence",
          "valueCode": "equivalent"
        },
        {
          "name": "concept",
          "valueCoding": {
            "system": "ToxSubstance ",
            "code": "249 ",
            "display": "6-ACETYLCODEINE"
          }
        }
      ]
    }
  ]
}
```

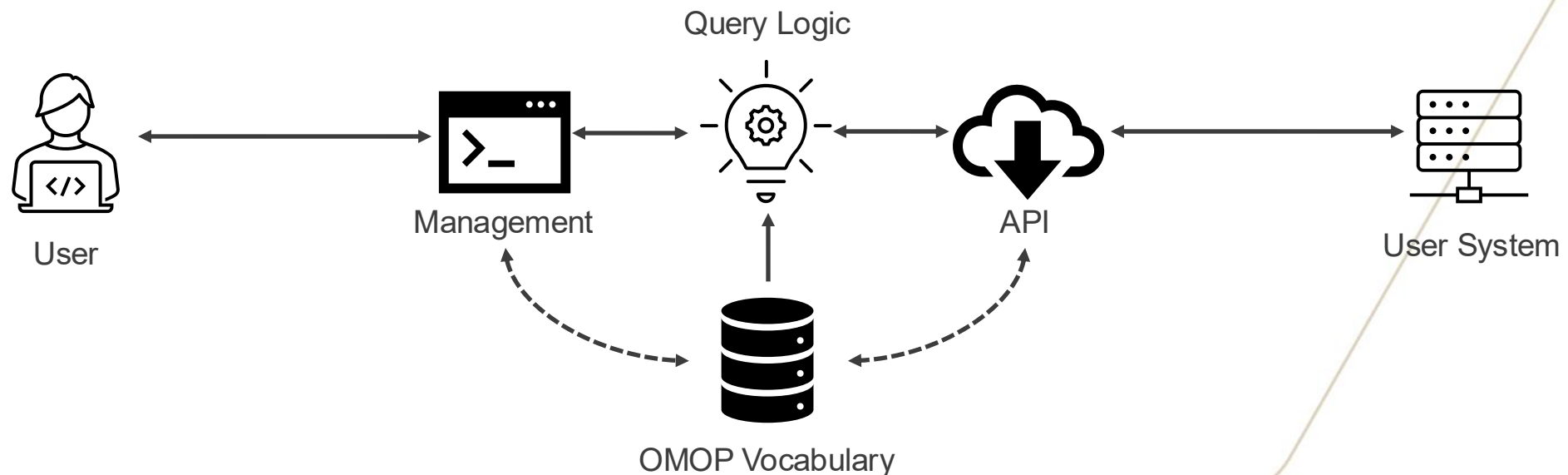
Preferred Substance Name Search Workflow

1. User enters their **substance name** (this can be automated as well).
2. Use "**Fuzzing**"-style approach to map the provided substance to a pseudo-standardized code from a desired system (ex: "mono acetyl morphine" to "6-MONOACETYLMORPHINE")
 - 1.(OPTIONAL) If relevant to users, the backend may at this point allow them to opt to select from a list of close matches to continue the process with.
3. Using the mapped code, check if it represents a preferred name or an alias for a preferred name.
4. Return the preferred **substance name** to use as an input for the system.

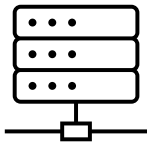


Future Terminology Server

- Current Proof of Concept Implementation
 - FHIR API for \$translate operation to return preferred substance(s)
 - Concept Manager
 - Add new substances and relationships
 - Search with Solr's basic query capability
- Core functionality & features
 - Intelligent Substance Name Suggestion API
 - Advanced Query Logic & Search Capabilities
 - Terminology Management & Update System



Use Case Example: Getting List of Preferred Substance Name



Abstractor Sandbox / System / ex: MOON

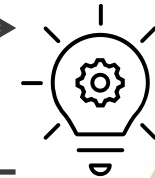


Terminology Service

From **Lab Report**:

Substance	Tested	Results
Morphine Free	Tested	Present

FHIR *\$translate* API with "Morphine Free" →



Translate with Query Logic

API response with list of preferred names →

- Free Morphine
- Morphine"



Choose substance name to use

Free Morphine

Morphine

*Toxicology terminology was developed based on the active substances list available in the National Violent Death Reporting System/State Unintentional Drug Overdose Reporting System

Terminology API Server

1. 6-ACETYLCODEINE's concept_code: [2000110223](#)
2. Vocabulary ID for substances: <system_defined_substance_id>
3. We want to get aliases of [6-ACETYLCODEINE](#) in <system_defined_substance> namespace
4. In FHIR, we can use \$translate operation API.

FHIR API for \$translate

```
GET [base]/ConceptMap/$translate?system=<vocab_id>&code=<concept_code>&targetsystem=<system_defined_substance>
System: <system_defined_substance>, code: 2000110223, targetsystem: <system_defined_substance>
```

The screenshot displays the HAPI FHIR web interface. On the left is a sidebar with a navigation menu. The top of the sidebar has tabs: (none), true, text, data, and count. Below these are sections for 'Server' (with a link to 'Server Home/Actions') and 'Resources' (listing ValueSet, AllergyIntolerance, CodeSystem, ConceptMap, Condition, Device, DeviceUseStatement, DocumentReference, Encounter, and Immunization). The main content area on the right shows server details in a table and a 'Server Actions' section with buttons for Conformance, History, and Transaction.

Server	HAPI FHIR
Software	OMOP v5.4 on FHIR R4 with MDI vocab - v1.6.2
FHIR Base	http://localhost:8080/fhir

Server Actions

Retrieve the server's **conformance** statement.
[Conformance](#)

Retrieve the update **history** across all resource types on the server.
[History](#)

Post a bundle containing multiple resources to the server and store all resources within a single atomic transaction.
[Transaction](#)

*Toxicology terminology was developed based on the active substances list available in the National Violent Death Reporting System/State Unintentional Drug Overdose Reporting System

Terminology API Client

The screenshot displays the Insomnia API client interface. The top bar shows the application name "Insomnia" and a search bar. The left sidebar contains a list of collections, including "Mortality Repo...", "Base Environment", "Manage Cookies (5)", and "Add Certificates". The main panel shows a GET request to the URL `http://localhost:8080/fhir/ConceptMap/$translate`. The request parameters are defined as follows:

Parameter	Value
system	<system_defined_substance_id>
code	2000110223
targetsystem	<system_defined_substance_id>

The response status is **200 OK** with a response time of 81 ms and a body size of 564 B. The response body is a JSON object:

```
{
  "resourceType": "Parameters",
  "parameter": [
    {
      "name": "result",
      "valueBoolean": true
    },
    {
      "name": "match",
      "part": [
        {
          "name": "equivalence",
          "valueCode": "equivalent"
        },
        {
          "name": "concept",
          "valueCoding": {
            "system": "<system_defined_substance_id>",
            "code": "2000110224",
            "display": "ACETYLCODEINE"
          }
        },
        {
          "name": "concept",
          "valueCoding": {
            "system": "<system_defined_substance_id>",
            "code": "2000110225",
            "display": "MONOACETYLCODEINE"
          }
        }
      ]
    }
  ]
}
```

Terminology Concept Management UI – search term

Concept Management

+ New Concept

+ New System

Search Term

System

Find Concept

Clear Search

Optional Columns: ☐ Domain ☐ Class

CONCEPT			START DATE	END DATE	MATCH SCORE	INVALID REASON
SYSTEM	CODE	NAME				
SNOMED	32485007	Hospital admission	Jan 31, 2002	Dec 31, 2099	1	
SNOMED	73425007	Inactive	Jan 31, 2002	Dec 31, 2099	1	
SNOMED	41228009	Obsolete	Jan 31, 2002	Dec 31, 2099	1	
SNOMED	261665005	UNKNOWN	Jan 1, 1970	Jan 30, 2013	1	Deleted
SNOMED	18632008	Patient status determination, deceased	Jan 31, 2002	Dec 31, 2099	1	
SNOMED	74964007	Other	Jan 31, 2002	Dec 31, 2099	1	
SNOMED	55561003	Active	Jan 31, 2002	Dec 31, 2099	1	

Selected Concept

Details

Inactive

Code

73425007

System

SNOMED

Domain

Meas Value

Class

Qualifier Value

Start Date

Jan 31, 2002

End Date

Dec 31, 2099

Edit

Hierarchy

Comments

No comments found

Comment Text*

Terminology Concept Management UI – concept management

 **Lab Vocabulary Manager**

Mapping Search

Create New Mapping

Concept Management

Message Viewer

Mapping Errors Viewer

Concept Management

Search Term

Optional Columns: ☒ Domain

SYSTEM	CODE
SNOMED	32485007
SNOMED	73425007
SNOMED	41228009
SNOMED	261665005
SNOMED	18632008
SNOMED	74964007
SNOMED	55561003

Version: 0.2.0

Create New Concept

Concept Name*

Concept Code*

System*

Concept Class*

Domain*

Start Date*

End Date*

Invalid Reason

 Cancel

 Save Concept



+ New Concept

+ New System

START DATE	END DATE	MATCH SCORE	INVALID REASON
Jan 31, 2002	Dec 31, 2099	1	
Jan 31, 2002	Dec 31, 2099	1	
Jan 31, 2002	Dec 31, 2099	1	
Jan 1, 1970	Jan 30, 2013	1	Deleted
Jan 31, 2002	Dec 31, 2099	1	
Jan 31, 2002	Dec 31, 2099	1	
Jan 31, 2002	Dec 31, 2099	1	